Author Correction: A dense ring of the trans-Neptunian object Quaoar outside its Roche limit

https://doi.org/10.1038/s41586-024-07031-w

Published online: 16 January 2024

Correction to: *Nature* https://doi.org/10.1038/s41586-022-05629-6

Published online 8 February 2023

Check for updates

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After re-examining our paper "A dense ring of the trans-Neptunian object Quaoar outside its Roche Limit", we have identified a miscalculation in the published optical depths for Quaoar's ring. The apparent optical depth of the ring must be corrected for geometrical effects and diffraction, as correctly presented in the "Data analysis" section of Methods. In particular, Airy diffraction by individual ring particles means that a narrow ring has half the apparent optical depth that is seen during Earth-based stellar occultations[1].

However, this correction by a factor of half has been wrongly applied twice in our chain of analysis, resulting in an underestimation by a factor of two in the actual normal optical depth and an underestimation of associated quantities such as the ring equivalent width and equivalent depth.

We highlight that the correction presented in this amendment does not alter the primary conclusion as outlined in the published paper. Meanwhile, these new values will permit a better assessment of the individual particle albedo when the ring photometry is available, as well as realistic collision rates in future N-body ring simulations.

In light of this correction, Table 1 and Extended Data Table 2 have been replaced in the article. For comparison, original versions of Table 1 and Extended Data Table 2 are available in the Supplementary Information associated with this amendment.

1. Cuzzi, J. N. Rings of Uranus: Not so thick, not so black. *Icarus* **63**, 312–316 (1985).

Supplementary information is available in the online version of this amendment.

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